Assignment: 2D Crystallographic Point Groups and Schoenflies Notation

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In 2D crystallography, there are exactly **10 crystallographic point groups**. These are restricted by the crystallographic restriction theorem, which states that a periodic lattice can only support 1-fold, 2-fold, 3-fold, 4-fold, and 6-fold rotational symmetries, along with mirror reflections.

To visualize these using the **"dice" analogy** discussed in class (where adding dots breaks the symmetry of a perfect cube), we place a maximum-symmetry geometric shape at each Bravais Lattice (BL) point. By systematically **"shading"** specific asymmetric sectors of these shapes we can reduce the symmetry to each of the 10 point groups.

Legend for all diagrams:

- ■ shaded sector (breaks or reduces symmetry)
- - - - mirror line
- • rotation-axis marker (circle = 2-fold; square = 4-fold; triangle = 3-fold; hexagon = 6-fold)

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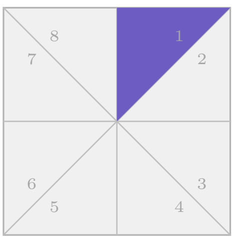
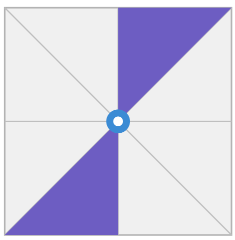
1 The Oblique Family (1-fold & 2-fold Rotations)

Base shape: A square divided into 8 identical right-triangle slices by cuts along the two diagonals and the horizontal/vertical mid-lines. Slices are numbered 1–8 clockwise from the top.



C_1 (HM: 1)
Identity only
Shade slice 1



C_2 (HM: 2)
180° rotation, no mirrors
Slices 1 & 5 (propeller)

Figure 1: Oblique family point groups.

- **Group 1 (C_1):** One shaded triangle breaks every rotational and mirror symmetry. Only the full 360° rotation returns the pattern to itself.

- **Group 2 (C_2):** Two diametrically opposite shaded triangles form a two-bladed propeller. Rotation by 180° swaps them; no mirror reflection maps the pattern onto itself.

2 The Rectangular Family (Mirrors & 2-fold Rotations)

C_s (HM: m)
1 mirror, no rotation
Slices 1 & 8

C_{2v} (HM: $2mm$)
180° + 2 mirrors
Slices 1,4,5,8 (rectangle)

Figure 2: Rectangular family point groups.

- **Group 3 (C_s):** Slices 1 and 8 are adjacent, forming a single shaded wedge above the horizontal midline. This creates exactly one mirror plane; no rotation (other than the trivial 360°) leaves the pattern invariant.

- **Group 4 (C_{2v}):** Two shaded wedges sit on opposite sides (right and left of the vertical midline), mimicking the symmetry of a non-square rectangle. The pattern is mapped onto itself by 180° rotation and by either of the two perpendicular mirror lines.

3 The Square Family (4-fold Rotations)

C_4 (HM: 4)
90° rotation, no mirrors
Slices 1,3,5,7 (pinwheel)

C_{4v} (HM: $4mm$)
90° + 4 mirrors
Unshaded (perfect square)

Figure 3: Square family point groups.

- **Group 5 (C_4):** Every alternate sector is shaded, producing a four-bladed pinwheel. Each 90° rotation maps shaded sectors onto shaded sectors. However, a mirror reflection reverses the "spin direction" of the pinwheel, so no mirror symmetry exists.
- **Group 6 (C_{4v}):** The unaltered square possesses the maximum possible symmetry for this family: four mirror lines (horizontal, vertical, and both diagonals) and four-fold rotational symmetry.

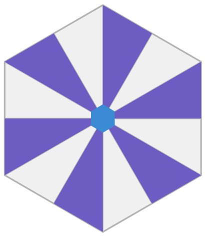
4 The Hexagonal Family (3-fold & 6-fold Rotations)

Base shape: A regular hexagon divided into 12 identical triangular slices by cuts from the centre to all 6 vertices *and* to all 6 edge midpoints. Slices are numbered 1–12 clockwise from the top.

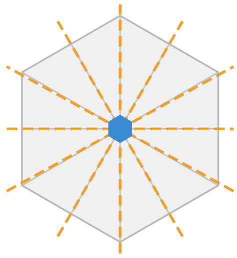
C_3 (HM: 3)
120° rotation, no mirrors
Slices 1,5,9 (3-blade pinwheel)

C_{3v} (HM: $3m$)
120° + 3 mirrors
Slices 1&12, 4&5, 8&9

Figure 4: Hexagonal family — 3-fold groups.



C_6 (HM: 6)
60° rotation, no mirrors
Slices 1,3,5,7,9,11 (6-blade pinwheel)



C_{6v} (HM: $6mm$)
60° + 6 mirrors
Unshaded (perfect hexagon)

Figure 5: Hexagonal family — 6-fold groups.

- **Group 7 (C_3):** Three sectors spaced 120° apart form a three-bladed pinwheel. Each 120° rotation maps blades onto blades; mirror reflections flip the handedness of the pinwheel, so there are no mirror planes.
- **Group 8 (C_{3v}):** Three *pairs* of adjacent sectors form arrowhead shapes, each pointing outward. A mirror line runs through the axis of each arrowhead. This is the symmetry of an equilateral triangle.
- **Group 9 (C_6):** Alternating shading gives a six-bladed pinwheel. Every 60° rotation maps blades onto blades, but mirrors flip the orientation of the alternating pattern.
- **Group 10 (C_{6v}):** The unaltered regular hexagon has six mirror lines (through opposite vertices and through mid-edge-to-mid-edge pairs) and six-fold rotational symmetry.

5 Tabular Summary

Point Group (HM)	**Schoenflies**	**Base Shape**	**Shading Strategy ("Dice" Analogy)**
1	C_1	Square (8 slices)	Shade 1 slice — completely asymmetric
m	C_s	Square (8 slices)	Shade 2 adjacent slices (1 mirror block)
2	C_2	Square (8 slices)	Shade 2 opposite slices (2-blade propeller)
$2mm$	C_{2v}	Square (8 slices)	Shade 2 blocks of 2 slices on opposite sides
4	C_4	Square (8 slices)	Shade 4 alternate slices (4-blade pinwheel)
$4mm$	C_{4v}	Square (8 slices)	Unshaded (perfect square)
3	C_3	Hexagon (12 slices)	Shade 3 slices spaced 120° apart (3-blade pinwheel)
$3m$	C_{3v}	Hexagon (12 slices)	Shade 3 adjacent pairs — arrowheads / equilateral triangle
6	C_6	Hexagon (12 slices)	Shade 6 alternate slices (6-blade pinwheel)
$6mm$	C_{6v}	Hexagon (12 slices)	Unshaded (perfect hexagon)

Table 1: Summary of 2D Crystallographic Point Groups